

Exact-Diameter Equivalence of Averaged and Singular Besov Energies for Ahlfors-Regular Measures on the Cube

Abstract

Let a Borel measure be Ahlfors regular through the scale $R = \sqrt{d}$, and suppose that its support lies in the Euclidean unit cube $[0, 1]^d$. We prove, with explicit positive finite structural constants, the equivalence between an averaged, ball-normalized quadratic energy and the corresponding singular Besov energy. All energies take values in the extended nonnegative reals, so the result requires no finiteness assumption on the energies. The exact cutoff $R = \sqrt{d}$ requires care because the balls are open: an open ball of radius equal to the cube diameter need not contain the opposite corner. We resolve this endpoint by a quantitative corner-avoidance construction. It produces a positive-measure anchor set at a radius strictly below R , which in turn controls the long-range part of the singular energy. The proof records the exact constants and the order in which they are selected. It is the human-readable counterpart of a Lean 4 and Mathlib formalization.

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1 Setting and conventions

Let

$$d \in \mathbb{N}, \quad d > 0, \quad X = \mathbb{R}^d, \quad Q_d = [0, 1]^d, \quad R = \sqrt{d}.$$

The space X carries its Euclidean metric and Borel sigma-algebra. Every ball $B(x, r)$ in this article is an *open* metric ball.

Let ν be a Borel measure on X , let

$$E = \text{supp } \nu,$$

and suppose

$$E \subseteq Q_d.$$

Fix real parameters

$$\alpha > 0, \quad s > 0, \quad s < \frac{\alpha}{2}, \quad 0 < c_A \leq C_A,$$

and put

$$p = \alpha + 2s.$$

In particular, $R > 0$, $C_A > 0$, and $p > 0$.

The Ahlfors hypotheses are

$$c_A r^\alpha \leq \nu(B(x, r)) \leq C_A r^\alpha \quad (x \in E, \ 0 < r \leq R). \quad (\text{AR})$$

All nonnegative integrals are interpreted as extended nonnegative Lebesgue integrals and take values in

$$\overline{\mathbb{R}}_+ = [0, \infty].$$

We write

$$\iota : [0, \infty) \longrightarrow \overline{\mathbb{R}}_+$$

for the canonical embedding. The extended inverse and product conventions are

$$0^{-1} = \infty, \quad \infty^{-1} = 0, \quad 0 \cdot \infty = \infty \cdot 0 = 0.$$

After the exact formal constants have been displayed, positive finite real numbers will be identified with their images under ι .

Let $u : X \rightarrow \mathbb{R}$ be Borel measurable and define

$$\Delta_u(x, y) = (u(x) - u(y))^2.$$

For $r \in \mathbb{R}$, set

$$D_r(x) = \int_{B(x,r)} \Delta_u(x,y) \, d\nu(y), \quad (1)$$

$$L_r = \int_X D_r(x) \, d\nu(x), \quad (2)$$

$$N_r = \int_X \nu(B(x,r))^{-1} D_r(x) \, d\nu(x). \quad (3)$$

The averaged energy is

$$\mathcal{A}_{s,R}(u) = \int_{(0,R]} r^{-1-2s} N_r \, dr, \quad (4)$$

and the unnormalized scale energy is

$$\mathcal{S}_{\alpha,s,R}(u) = \int_{(0,R]} r^{-1-\alpha-2s} L_r \, dr = \int_{(0,R]} r^{-1-p} L_r \, dr. \quad (5)$$

The singular integrand is defined explicitly on the diagonal:

$$\Phi_{p,u}(x,y) = \begin{cases} 0, & x = y, \\ \Delta_u(x,y)|x-y|^{-p}, & x \neq y. \end{cases}$$

The singular Besov energy and global quadratic variation are

$$\mathcal{B}_{\alpha,s}(u) = \int_X \int_X \Phi_{p,u}(x,y) \, d\nu(y) \, d\nu(x), \quad (6)$$

$$\mathcal{V}(u) = \int_X \int_X \Delta_u(x,y) \, d\nu(y) \, d\nu(x). \quad (7)$$

The diagonal convention in (6) is part of the definition; the singular energy is not defined by silently evaluating a product $0 \cdot \infty$.

Remark 1.1 (The function space represented by the formal theorem). The formal theorem quantifies over globally Borel measurable functions $u : X \rightarrow \mathbb{R}$. It therefore applies, in particular, to any globally measurable representative of an $L^2(E, \nu)$ function. It is not a theorem stated on an L^2 quotient type, and it does not formalize the selection or extension of a representative defined only on E .

2 Exact structural constants and the main theorem

Let

$$\mathcal{C}_d = \{0,1\}^d$$

be the set of cube corners. Since $\alpha > 0$, one can choose, once and for all and before choosing ν or u , a number $\delta > 0$ such that

$$2\delta \leq R \quad (8)$$

and

$$\sum_{v \in \mathcal{C}_d} C_A(2\delta)^\alpha < c_A R^\alpha. \quad (9)$$

Equivalently, the left side is $2^d C_A(2\delta)^\alpha$.

Define the real geometric quantities

$$\rho_\delta = \sqrt{d - \delta^2}, \quad (10)$$

$$\varepsilon_\delta = \frac{R - \rho_\delta}{2}, \quad (11)$$

$$\lambda_\delta = \frac{R + \rho_\delta}{2R}, \quad (12)$$

$$r_\delta = \lambda_\delta R = \frac{R + \rho_\delta}{2}. \quad (13)$$

The exact constants used by the formal theorem are the following elements of $\overline{\mathbb{R}}_+$:

$$m_- = \iota(c_A \varepsilon_\delta^\alpha), \quad (14)$$

$$m_+ = \iota(C_A R^\alpha), \quad (15)$$

$$K_\delta = \iota \left(\frac{r_\delta^{-p} - R^{-p}}{p} \right)^{-1}, \quad (16)$$

$$G_\delta = \left(\frac{4m_+}{m_-} \right) K_\delta, \quad (17)$$

$$D_\delta = (\iota(p) + \iota(R)^{-p} G_\delta) \iota(C_A), \quad (18)$$

$$c = D_\delta^{-1}, \quad (19)$$

$$C = \iota(c_A)^{-1} \iota(p)^{-1}. \quad (20)$$

Here $\iota(R)^{-p}$ denotes the extended nonnegative real power. Since all of the relevant real quantities are positive and finite, it agrees with $\iota(R^{-p})$.

Theorem 2.1 (Exact-diameter equivalence). *Fix d, α, s, c_A, C_A as above, choose a structural δ satisfying (8)–(9), and define c, C by (10)–(20). Then*

$$0 < c < \infty, \quad 0 < C < \infty.$$

For every Borel measure ν on X and every globally Borel measurable $u : X \rightarrow \mathbb{R}$, if ν satisfies (AR) and $\text{supp } \nu \subseteq Q_d$, then

$$c \mathcal{B}_{\alpha, s}(u) \leq \mathcal{A}_{s, R}(u) \leq C \mathcal{B}_{\alpha, s}(u). \quad (21)$$

The constants are selected in the order

$$(d, \alpha, s, c_A, C_A) \longmapsto (\delta, c, C) \longmapsto (\nu, u).$$

In particular, c and C do not depend on ν or u .

The remainder of the article proves Theorem 2.1.

3 Measure-theoretic preliminaries

Lemma 3.1 (Automatic finiteness and sigma-finiteness). *Under the hypotheses of Theorem 2.1, the measure ν is finite. Consequently it is sigma-finite.*

Proof. The support $E = \text{supp } \nu$ is closed and lies in the compact cube Q_d , hence E is compact. For every $x \in E$, the upper Ahlfors estimate at radius R gives

$$\nu(B(x, R)) \leq C_A R^\alpha < \infty.$$

The balls $B(x, R)$, $x \in E$, form an open cover of E . Compactness gives a finite subcover, so finite subadditivity yields $\nu(E) < \infty$. The complement of the support is ν -null in the second-countable Euclidean space. Therefore

$$\nu(X) = \nu(E) < \infty.$$

A finite measure is sigma-finite. □

Lemma 3.2 (Measurability of the energy kernels). *If $u : X \rightarrow \mathbb{R}$ is Borel measurable, then all integrands in (1)–(7) are measurable. The maps $x \mapsto D_r(x)$, $r \mapsto L_r$, and $r \mapsto N_r$ are measurable wherever they are used. All applications of Tonelli below are therefore valid.*

Proof. The map

$$(x, y) \mapsto \Delta_u(x, y)$$

is Borel measurable because subtraction and squaring are continuous. The distance map is continuous, and hence

$$\{(r, x, y) \in \mathbb{R} \times X \times X : |x - y| < r\}$$

is Borel. Thus

$$(r, x, y) \mapsto \mathbf{1}_{\{|x-y|<r\}} \Delta_u(x, y)$$

is a nonnegative measurable function. Parameterized Tonelli, using the sigma-finiteness from Lemma 3.1, shows that its integrals in y , and subsequently in x , are measurable functions of the remaining variables. Applying the same argument to the indicator alone shows that $(r, x) \mapsto \nu(B(x, r))$ is measurable. Extended inversion, multiplication, and real power are Borel operations on $\overline{\mathbb{R}}_+$. This proves the assertions for D_r , L_r , N_r , and the scale integrands. The kernels in (6)–(7) are measurable by the same closure properties, with the diagonal handled by a measurable piecewise definition. □

Lemma 3.3 (Atomlessness). *Every singleton has zero ν -measure.*

Proof. First let $x \in E$. For every $0 < r \leq R$,

$$\nu(\{x\}) \leq \nu(B(x, r)) \leq C_A r^\alpha.$$

Since $\alpha > 0$, the right side tends to zero as $r \downarrow 0$, and therefore $\nu(\{x\}) = 0$. If $x \notin E$, the definition of support provides an open neighborhood U of x with $\nu(U) = 0$, whence $\nu(\{x\}) = 0$ as well. □

Remark 3.4. Lemma 3.3 is independently useful, but the formal definition of $\mathcal{B}_{\alpha,s}$ does not rely on it: the diagonal integrand is set to zero explicitly.

4 Normalization by Ahlfors regularity

Lemma 4.1 (Fixed-scale normalization). *For $0 < r \leq R$,*

$$(C_A r^\alpha)^{-1} L_r \leq N_r \leq (c_A r^\alpha)^{-1} L_r.$$

Proof. The support is conull. Thus for almost every x , $x \in E$, and (AR) gives

$$c_A r^\alpha \leq \nu(B(x, r)) \leq C_A r^\alpha.$$

All three quantities are positive and finite. Extended inversion reverses the inequalities, so

$$(C_A r^\alpha)^{-1} \leq \nu(B(x, r))^{-1} \leq (c_A r^\alpha)^{-1}.$$

Multiplication by $D_r(x) \in \overline{\mathbb{R}}_+$ preserves the order. Integration in x proves the result. $\square$

Lemma 4.2 (Integrated normalization). *One has*

$$C_A^{-1} \mathcal{S}_{\alpha, s, R}(u) \leq \mathcal{A}_{s, R}(u) \leq c_A^{-1} \mathcal{S}_{\alpha, s, R}(u). \quad (22)$$

In particular,

$$\mathcal{S}_{\alpha, s, R}(u) \leq C_A \mathcal{A}_{s, R}(u). \quad (23)$$

Proof. For $A > 0$ and $r > 0$,

$$r^{-1-2s} (A r^\alpha)^{-1} = A^{-1} r^{-1-\alpha-2s}.$$

Multiply the inequalities of Lemma 4.1 by r^{-1-2s} , integrate over $(0, R]$, and use (4)–(5). This gives (22). Since C_A is positive and finite, multiplication by C_A and cancellation of $C_A C_A^{-1} = 1$ yield (23). The cancellation is valid in $\overline{\mathbb{R}}_+$ because the cancelled factor is neither zero nor infinity. $\square$

5 The scale integral and its exact Tonelli kernel

Define

$$K_{p, R}(t) = \begin{cases} \frac{t^{-p} - R^{-p}}{p}, & 0 < t \leq R, \\ 0, & \text{otherwise.} \end{cases} \quad (24)$$

The associated truncated double energy is

$$\mathcal{T}_{p, R}(u) = \int_X \int_X K_{p, R}(|x - y|) \Delta_u(x, y) \, d\nu(y) \, d\nu(x). \quad (25)$$

Lemma 5.1 (Scalar scale integral). *If $p > 0$ and $0 < t \leq R$, then*

$$\int_{(t, R]} r^{-1-p} \, dr = \frac{t^{-p} - R^{-p}}{p}.$$

Proof. The function $-p^{-1} r^{-p}$ is an antiderivative of r^{-1-p} on $(0, \infty)$. Evaluation at t and R gives the formula. The integrand is integrable on $(t, R]$ because the interval is bounded away from zero. $\square$

Lemma 5.2 (Exact Tonelli identity). *For every measurable $u : X \rightarrow \mathbb{R}$,*

$$\mathcal{S}_{\alpha,s,R}(u) = \mathcal{T}_{p,R}(u). \quad (26)$$

Proof. Expand L_r from (2) and (1):

$$\mathcal{S}_{\alpha,s,R}(u) = \int_{(0,R]} \int_X \int_X \mathbf{1}_{\{|x-y|<r\}} r^{-1-p} \Delta_u(x,y) \, d\nu(y) \, d\nu(x) \, dr.$$

The integrand is nonnegative and measurable by Lemma 3.2, so Tonelli permits both changes of order. Fix x, y , and put $t = |x - y|$.

If $x \neq y$, then $t > 0$. For $0 < t \leq R$, the allowed scales are exactly $(t, R]$, and Lemma 5.1 gives

$$\int_{(0,R]} \mathbf{1}_{\{t < r\}} r^{-1-p} \, dr = K_{p,R}(t).$$

If $t > R$, there is no allowed scale and both sides are zero. If $t = R$, there is likewise no $r \in (0, R]$ with $t < r$; this is precisely why $K_{p,R}(R) = 0$.

If $x = y$, then $\Delta_u(x, x) = 0$, so the complete scale integrand is identically zero. We handle this case before factoring the scalar integral: the unmultiplied integral of r^{-1-p} down to zero diverges, and no informal infinity-times-zero calculation is used.

After integration in x and y , the resulting expression is exactly (25). $\square$

6 Comparison with the full singular kernel

Lemma 6.1 (Pointwise kernel decomposition). *For $t > 0$,*

$$t^{-p} \leq pK_{p,R}(t) + R^{-p}. \quad (27)$$

Equality holds whenever $0 < t \leq R$.

Proof. If $0 < t \leq R$, the definition gives

$$pK_{p,R}(t) + R^{-p} = t^{-p} - R^{-p} + R^{-p} = t^{-p}.$$

If $t > R$, then $K_{p,R}(t) = 0$. Since $p > 0$, the function $t \mapsto t^{-p}$ is decreasing on $(0, \infty)$, and hence $t^{-p} \leq R^{-p}$. $\square$

Lemma 6.2 (Full-energy decomposition). *One has*

$$\mathcal{B}_{\alpha,s}(u) \leq p\mathcal{T}_{p,R}(u) + R^{-p}\mathcal{V}(u). \quad (28)$$

Moreover, equality holds if $|x - y| \leq R$ for $\nu \otimes \nu$ -almost every pair (x, y) . In particular, equality holds under the cube-support hypothesis.

Proof. On the diagonal, both $\Phi_{p,u}(x, x)$ and $\Delta_u(x, x)$ are zero. Off the diagonal, multiply (27) by $\Delta_u(x, y)$, and integrate twice. Tonelli and distributivity for nonnegative extended integrals give (28).

If almost every pair has distance at most R , then the equality case of Lemma 6.1 applies almost everywhere. If $E \subseteq Q_d$, then

$$|x - y|^2 = \sum_{j=1}^d (x_j - y_j)^2 \leq d$$

for $x, y \in E$. Since E is conull, this proves the final assertion. Pairs at exact distance R are included: their truncated-kernel contribution is zero, while their contribution appears in the $R^{-p}\mathcal{V}(u)$ term. $\square$

Lemma 6.3 (Truncated energy is bounded by full energy). *One has*

$$\mathcal{T}_{p,R}(u) \leq p^{-1}\mathcal{B}_{\alpha,s}(u). \quad (29)$$

Proof. For $0 < t \leq R$,

$$K_{p,R}(t) = \frac{t^{-p} - R^{-p}}{p} \leq p^{-1}t^{-p}.$$

For $t > R$, the left side is zero. On the diagonal, the complete integrands are both defined to be zero. Multiplication by $\Delta_u(x, y)$ and two applications of monotonicity of the nonnegative integral prove (29). $\square$

Corollary 6.4 (The easy comparison).

$$\mathcal{A}_{s,R}(u) \leq c_A^{-1}p^{-1}\mathcal{B}_{\alpha,s}(u).$$

Proof. By Lemmas 4.2, 5.2, and 6.3,

$$\mathcal{A}_{s,R}(u) \leq c_A^{-1}\mathcal{S}_{\alpha,s,R}(u) = c_A^{-1}\mathcal{T}_{p,R}(u) \leq c_A^{-1}p^{-1}\mathcal{B}_{\alpha,s}(u).$$

$\square$

7 A structural point away from every corner

Lemma 7.1 (Existence of the structural corner scale). *A number $\delta > 0$ satisfying (8)–(9) exists and depends only on d, α, c_A, C_A .*

Proof. Since $\alpha > 0$,

$$(2\delta)^\alpha \longrightarrow 0 \quad (\delta \downarrow 0).$$

There are 2^d corners, so

$$\sum_{v \in \mathcal{C}_d} C_A(2\delta)^\alpha = 2^d C_A(2\delta)^\alpha \longrightarrow 0.$$

The target $c_A R^\alpha$ is strictly positive. Thus (9) holds for all sufficiently small positive δ . Shrinking δ further ensures $2\delta \leq R$. $\square$

Lemma 7.2 (Corner-neighborhood estimate). *For every corner $v \in \mathcal{C}_d$,*

$$\nu(B(v, \delta)) \leq C_A(2\delta)^\alpha. \quad (30)$$

Consequently, for

$$\mathcal{N}_\delta = \bigcup_{v \in \mathcal{C}_d} B(v, \delta),$$

one has

$$\nu(\mathcal{N}_\delta) \leq \sum_{v \in \mathcal{C}_d} C_A(2\delta)^\alpha < c_A R^\alpha. \quad (31)$$

Proof. Fix v . If $\nu(B(v, \delta)) = 0$, then (30) is immediate. Otherwise the ball meets the support: choose $y_v \in B(v, \delta) \cap E$. Indeed, the complement of E is null, so a positive-measure Borel set cannot be contained in it. If $z \in B(v, \delta)$, then

$$|z - y_v| \leq |z - v| + |v - y_v| < 2\delta.$$

Thus

$$B(v, \delta) \subseteq B(y_v, 2\delta).$$

Since $y_v \in E$, $2\delta > 0$, and $2\delta \leq R$, the upper Ahlfors estimate gives

$$\nu(B(v, \delta)) \leq \nu(B(y_v, 2\delta)) \leq C_A(2\delta)^\alpha.$$

Finite subadditivity over the 2^d corners, followed by (9), proves (31). $\square$

Lemma 7.3 (Supported point uniformly away from the corners). *Assume $\nu \neq 0$. Then there exists $q \in E$ such that*

$$\delta \leq |q - v| \quad (v \in \mathcal{C}_d). \quad (32)$$

Proof. Since $\nu \neq 0$, the support is nonempty; choose $x_0 \in E$. The lower Ahlfors estimate at radius R gives

$$c_A R^\alpha \leq \nu(B(x_0, R)) \leq \nu(X). \quad (33)$$

If every supported point belonged to $\mathcal{N}_\delta$, then conullity of E would imply

$$\nu(X) = \nu(E) \leq \nu(\mathcal{N}_\delta).$$

This contradicts (31) and (33). Hence some

$$q \in E \setminus \mathcal{N}_\delta$$

exists. Not belonging to the open ball $B(v, \delta)$ is exactly $|q - v| \geq \delta$, proving (32). $\square$

8 Subdiametral geometry and the anchor set

Lemma 8.1 (Quantitative subdiametral radius). *Let $q \in Q_d$ satisfy (32). Then*

$$0 \leq \rho_\delta < R$$

and

$$|x - q| \leq \rho_\delta \quad (x \in Q_d). \quad (34)$$

Proof. Define a corner $a \in \mathcal{C}_d$ coordinatewise by

$$a_j = \begin{cases} 1, & q_j > \frac{1}{2}, \\ 0, & q_j \leq \frac{1}{2}. \end{cases}$$

For every $x \in Q_d$ and every coordinate j ,

$$|q_j - a_j|^2 + |x_j - q_j|^2 \leq 1. \quad (35)$$

Indeed, if $q_j > 1/2$, then $a_j = 1$ and

$$|x_j - q_j| \leq q_j, \quad (1 - q_j)^2 + q_j^2 \leq 1.$$

If $q_j \leq 1/2$, then $a_j = 0$ and

$$|x_j - q_j| \leq 1 - q_j, \quad q_j^2 + (1 - q_j)^2 \leq 1.$$

Summing (35) gives

$$|q - a|^2 + |x - q|^2 \leq d.$$

By (32), $\delta \leq |q - a|$, and therefore

$$|x - q|^2 \leq d - |q - a|^2 \leq d - \delta^2.$$

This proves (34). It also shows $d - \delta^2 \geq 0$. Since $\delta > 0$,

$$d - \delta^2 < d,$$

and monotonicity of the square root gives

$$\rho_\delta = \sqrt{d - \delta^2} < \sqrt{d} = R.$$

□

Lemma 8.2 (Positive-measure anchor set below the diameter). *Assume $\nu \neq 0$, and choose q as in Lemma 7.3. Let*

$$S = B(q, \varepsilon_\delta) \cap E.$$

Then S is measurable, $S \subseteq E$, and

$$m_- \leq \nu(S), \quad (36)$$

so in particular $\nu(S) > 0$. Moreover,

$$|x - z| < r_\delta \quad (x \in E, z \in S), \quad (37)$$

where

$$0 < r_\delta < R.$$

Proof. By Lemma 8.1,

$$0 \leq \rho_\delta < R.$$

Hence

$$\varepsilon_\delta = \frac{R - \rho_\delta}{2} > 0, \quad r_\delta = \frac{R + \rho_\delta}{2} < R.$$

Also $\varepsilon_\delta \leq R$. The open ball is Borel, the support is closed, and therefore S is measurable. Since E is conull,

$$\nu(S) = \nu(B(q, \varepsilon_\delta)).$$

The point q belongs to E , so the lower Ahlfors estimate gives

$$\nu(S) \geq c_A \varepsilon_\delta^\alpha = m_-.$$

This quantity is strictly positive.

For $x \in E \subseteq Q_d$ and $z \in S$, use Lemma 8.1 and the strict ball membership of z :

$$|x - z| \leq |x - q| + |q - z| < \rho_\delta + \varepsilon_\delta = \frac{R + \rho_\delta}{2} = r_\delta.$$

This proves (37). □

Lemma 8.3 (Total-mass upper bound). *Under the assumptions of Lemma 8.2,*

$$\nu(X) \leq m_+. \tag{38}$$

Proof. Because $\nu(S) > 0$, the set S is nonempty; choose $z \in S$. Then $z \in E$. By (37) and $r_\delta < R$,

$$E \subseteq B(z, R).$$

This containment is strict at the metric level: every supported point has distance less than r_δ , not merely at most R . Since E is conull,

$$\nu(X) \leq \nu(B(z, R)) \leq C_A R^\alpha = m_+,$$

where the second inequality is the upper Ahlfors estimate at z and radius R . □

Remark 8.4 (Why the corner construction is needed). It is false in general that an open ball $B(x, R)$, with $R = \sqrt{d}$, contains the whole cube: opposite corners are exactly distance R apart. Lemma 8.2 replaces this invalid endpoint argument by a structural scale $r_\delta < R$. The strict gap simultaneously supplies an open-ball covering in Lemma 8.3 and a positive terminal interval of scales below.

9 The anchor estimate for global variation

Lemma 9.1 (Three-point squared-difference inequality). *For all $x, y, z \in X$,*

$$\Delta_u(x, y) \leq 2\Delta_u(x, z) + 2\Delta_u(z, y). \tag{39}$$

Proof. Write

$$u(x) - u(y) = (u(x) - u(z)) + (u(z) - u(y)).$$

The elementary inequality $(a + b)^2 \leq 2a^2 + 2b^2$ gives (39). □

Lemma 9.2 (Anchor control of global variation). *Assume $\nu \neq 0$, and let S be the anchor set of Lemma 8.2. Then*

$$\mathcal{V}(u) \leq \frac{4m_+}{m_-} L_{r_\delta}. \tag{40}$$

Proof. Define

$$H(x) = \int_S \Delta_u(x, z) \, d\nu(z).$$

Integrate (39) over $z \in S$. Symmetry of Δ_u gives

$$\nu(S)\Delta_u(x, y) \leq 2H(x) + 2H(y). \quad (41)$$

Integrating in x and y and using Tonelli yields

$$\begin{aligned} \nu(S)\mathcal{V}(u) &\leq 2\nu(X) \int_X H(x) \, d\nu(x) + 2\nu(X) \int_X H(y) \, d\nu(y) \\ &= 4\nu(X) \int_X H(x) \, d\nu(x). \end{aligned}$$

For almost every x , one has $x \in E$. By (37), $S \subseteq B(x, r_\delta)$, and hence

$$H(x) \leq \int_{B(x, r_\delta)} \Delta_u(x, z) \, d\nu(z) = D_{r_\delta}(x).$$

It follows that

$$\nu(S)\mathcal{V}(u) \leq 4\nu(X)L_{r_\delta}. \quad (42)$$

By (36) and (38),

$$0 < m_- \leq \nu(S) \leq \nu(X) \leq m_+ < \infty.$$

Thus division by $\nu(S)$ is legitimate in $\overline{\mathbb{R}}_+$, and

$$\mathcal{V}(u) \leq \frac{4\nu(X)}{\nu(S)}L_{r_\delta} \leq \frac{4m_+}{m_-}L_{r_\delta}.$$

□

10 Recovering one local scale from the scale integral

Lemma 10.1 (Monotonicity of local energy). *The map $r \mapsto L_r$ is nondecreasing.*

Proof. If $r \leq t$, then $B(x, r) \subseteq B(x, t)$ for every x . Since $\Delta_u \geq 0$,

$$D_r(x) \leq D_t(x).$$

Integration in x gives $L_r \leq L_t$. □

Lemma 10.2 (Terminal scale interval). *Under the assumptions of Lemma 8.2,*

$$L_{r_\delta} \leq K_\delta \mathcal{S}_{\alpha, s, R}(u). \quad (43)$$

Proof. Because $0 < r_\delta < R$ and $p > 0$,

$$k_\delta := \int_{(r_\delta, R]} r^{-1-p} \, dr = \frac{r_\delta^{-p} - R^{-p}}{p} > 0. \quad (44)$$

For $r \in (r_\delta, R]$, Lemma 10.1 gives $L_{r_\delta} \leq L_r$. Therefore

$$k_\delta L_{r_\delta} \leq \int_{(r_\delta, R]} r^{-1-p} L_r \, dr \leq \mathcal{S}_{\alpha, s, R}(u).$$

The number k_δ is positive and finite, so extended cancellation gives

$$L_{r_\delta} \leq k_\delta^{-1} \mathcal{S}_{\alpha,s,R}(u) = K_\delta \mathcal{S}_{\alpha,s,R}(u).$$

□

Corollary 10.3 (Structural control of global variation). *If $\nu \neq 0$, then*

$$\mathcal{V}(u) \leq G_\delta \mathcal{S}_{\alpha,s,R}(u) = G_\delta \mathcal{T}_{p,R}(u). \quad (45)$$

Proof. Combine Lemmas 9.2 and 10.2:

$$\mathcal{V}(u) \leq \frac{4m_+}{m_-} L_{r_\delta} \leq \frac{4m_+}{m_-} K_\delta \mathcal{S}_{\alpha,s,R}(u).$$

The coefficient is G_δ by (17), and Lemma 5.2 gives the final equality. □

11 Proof of the main theorem

Lemma 11.1 (Positivity and finiteness of the constants). *The constants in (14)–(20) satisfy*

$$0 < c < \infty, \quad 0 < C < \infty.$$

Proof. From $d > 0$, $R = \sqrt{d} > 0$. From (8), $\delta \leq R/2 < R$, and hence

$$0 \leq \rho_\delta = \sqrt{d - \delta^2} < R.$$

Therefore

$$\varepsilon_\delta > 0, \quad 0 < r_\delta < R.$$

Since $\alpha, c_A, C_A, R, \varepsilon_\delta$ are positive,

$$0 < m_- < \infty, \quad 0 < m_+ < \infty.$$

Because $p > 0$ and $r_\delta < R$, the negative power is strictly decreasing:

$$r_\delta^{-p} > R^{-p}.$$

Thus the real number in parentheses in (16) is positive and finite, and consequently

$$0 < K_\delta < \infty.$$

It follows successively from (17) and (18) that

$$0 < G_\delta < \infty, \quad 0 < D_\delta < \infty.$$

The inverse of a positive finite extended nonnegative real is again positive and finite, proving the claim for $c = D_\delta^{-1}$. The same argument applied to c_A and p proves the claim for $C = c_A^{-1} p^{-1}$. □

Proof of Theorem 2.1. The constants δ, c, C have already been selected solely from the structural data

$$d, \quad \alpha, \quad s, \quad c_A, \quad C_A.$$

Lemma 11.1 proves their positivity and finiteness. Now fix an arbitrary qualifying measure ν and measurable function u .

The upper inequality follows for every ν , including the zero measure, from Corollary 6.4:

$$\mathcal{A}_{s,R}(u) \leq c_A^{-1} p^{-1} \mathcal{B}_{\alpha,s}(u) = C \mathcal{B}_{\alpha,s}(u).$$

For the reverse inequality, first suppose $\nu = 0$. Then every ball energy, local energy, scale energy, truncated energy, global variation, and singular energy is zero. In the normalized local energy, $\nu(B(x, r))^{-1} = \infty$ and $D_r(x) = 0$; the stipulated $\mathbb{R}_+$ convention gives $\infty \cdot 0 = 0$. Thus $\mathcal{A}_{s,R}(u) = 0$, and the reverse inequality is $0 \leq 0$.

Suppose now that $\nu \neq 0$. Corollary 10.3 and the full-energy decomposition give

$$\begin{aligned} \mathcal{B}_{\alpha,s}(u) &\leq p \mathcal{T}_{p,R}(u) + R^{-p} \mathcal{V}(u) \\ &\leq (p + R^{-p} G_\delta) \mathcal{T}_{p,R}(u). \end{aligned}$$

Using the Tonelli identity and the lower side of the normalization comparison in its rearranged form (23),

$$\begin{aligned} \mathcal{B}_{\alpha,s}(u) &\leq (p + R^{-p} G_\delta) \mathcal{S}_{\alpha,s,R}(u) \\ &\leq (p + R^{-p} G_\delta) C_A \mathcal{A}_{s,R}(u) \\ &= D_\delta \mathcal{A}_{s,R}(u). \end{aligned}$$

The coefficient D_δ is positive and finite. Hence extended cancellation is valid, and

$$D_\delta^{-1} \mathcal{B}_{\alpha,s}(u) \leq \mathcal{A}_{s,R}(u).$$

Since $c = D_\delta^{-1}$, this is the required reverse inequality. $\square$

Remark 11.2 (The unused exponent restriction). The proof uses only $p = \alpha + 2s > 0$, which follows from $\alpha > 0$ and $s > 0$. The restriction $s < \alpha/2$ remains in Theorem 2.1 to match the formal target, but it is not needed for the comparison proved here.

12 Endpoint and extended-valued safeguards

For clarity, the proof uses the following safeguards explicitly.

1. The scale domain is $(0, R]$, while the metric balls are open. Thus a pair at distance R is in no ball at an allowed scale. Accordingly $K_{p,R}(R) = 0$.
2. The missing contribution at distance R is not discarded. It appears exactly in the $R^{-p} \mathcal{V}(u)$ term of the full-energy decomposition.
3. The diagonal is handled before the scalar scale integral is factored. The scalar integral

$$\int_0^R r^{-1-p} \, dr = \infty$$

diverges, but the complete diagonal integrand is identically zero because $\Delta_u(x, x) = 0$.

4. The cube diameter estimate gives only $|x - y| \leq R$. It does not imply that an open R -ball centered at an arbitrary cube point contains the cube. The corner-avoidance construction produces the strict anchor scale $r_\delta < R$.
5. The zero measure satisfies the formal Ahlfors predicate vacuously, because its support is empty. It is therefore a genuine case of the theorem and is treated separately.
6. No subtraction of extended energies is used. The only subtraction occurs between finite positive real powers in $K_{p,R}$ and k_δ , before embedding into $\overline{\mathbb{R}}_+$.
7. Every cancellation or division in $\overline{\mathbb{R}}_+$ is made only after the relevant coefficient has been proved positive and finite.

A Correspondence with the Lean 4 formalization

The compiled final declaration is

`BesovVerification.exists_exactCube_besovEnergy_equivalence_constants.`

Its quantifier order is:

$$\forall d, \alpha, s, c_A, C_A, \quad \exists c, C, \quad \forall \nu, u.$$

It assumes

$$d > 0, \quad \alpha > 0, \quad s > 0, \quad s < \alpha/2, \quad c_A > 0, \quad c_A \leq C_A,$$

and concludes the two inequalities in (21), with $0 < c, C < \infty$.

Lean source file	Mathematical role in this article
<code>Definitions.lean</code>	Ambient Euclidean space, unit cube, Ahlfors predicate, ball, local, normalized, averaged, singular, and global energies.
<code>Measurability.lean</code>	Parameterized measurability for moving balls, ball measures, local energies, and normalized scale integrands; supports Lemma 3.2.
<code>Ahlfors.lean</code>	Atomlessness from the upper Ahlfors estimate; corresponds to Lemma 3.3.
<code>CoreLemmas.lean</code>	The three-point squared-difference inequality and monotonicity of local energy; Lemmas 9.1 and 10.1.
<code>ScaleIntegral.lean</code>	The real and extended-nonnegative scalar integral in Lemma 5.1.
<code>Normalization.lean</code>	Fixed-scale and integrated Ahlfors normalization; Lemmas 4.1 and 4.2.
<code>EnergyIdentity.lean</code>	Measurable triple kernel, diagonal handling, Tonelli swaps, and the exact identity (26).
<code>MainTheorem.lean</code>	Comparison of the truncated kernel with the singular kernel and the easy upper inequality; Lemma 6.3 and Corollary 6.4.
<code>FullEnergyDecomposition.lean</code>	The pointwise singular-kernel identity and the global-variation tail; Lemmas 6.1 and 6.2.
<code>CubeAnchor.lean</code>	Corner neighborhoods, structural corner-scale existence, compact-support finiteness, corner avoidance, subdiametral geometry, and quantitative anchor balls; the compact-support, corner-avoidance, and anchor-geometry arguments above.
<code>FixedCornerScaleAnchor.lean</code>	Packages the anchor construction with δ fixed before the measure, which is essential for the final quantifier order.
<code>AnchorEstimate.lean</code>	Integration of the three-point inequality over the anchor set; Lemma 9.2.
<code>StructuralAnchorEstimate.lean</code>	Replaces the actual anchor and total masses by the uniform bounds m_- and m_+ .
<code>TailEstimate.lean</code>	Recovers L_{r_δ} from the terminal scale interval; Lemma 10.2.
<code>AnchorTailComparison.lean</code>	Composes anchor, terminal-scale, and normalization estimates.
<code>ExactCubeConstants.lean</code>	Defines $\rho_\delta, \varepsilon_\delta, \lambda_\delta, r_\delta$ and the exact constants (14)–(20), and proves their positivity and finiteness.
<code>ExactCubeMainTheorem.lean</code>	Handles the zero measure, assembles both inequalities, and establishes the final structural quantifier order.

B Pinned verification environment

The formal development was compiled in the following pinned environment:

Component	Version
Lean	4.32.1
Lean commit	f054605aea4b840552cca2e725580bffd1e1b704
Lake	5.0.0
Elan	4.2.3
Toolchain	leanprover/lean4:v4.32.1
Mathlib commit	520045ab14e26149ee970e2e617ca04b09bde5d6

The final theorem's reported logical dependencies are

`propext`, `Classical.choice`, `Quot.sound`.

These are standard Lean axioms used by Mathlib. The checked project source contains no unfinished proof term and introduces no custom axiom asserting the mathematical result.